

TRIBHUVAN UNIVERSITY
INSTITUTE OF SCIENCE AND TECHNOLOGY



A Final year internship report as “Python Developer” at Docsumo

In partial fulfillment of the requirement for the Bachelor of Science in Computer Science and
Information Technology

Under the supervision of

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SUPERVISOR'S RECOMMENDATION

I hereby recommend that the report prepared under my supervision by Mr. Hemanta Pokharel (TU Exam Roll No. X) entitled “Python Developer” at “Docsumo” (Megabit Services Pvt. Ltd.) in partial fulfillment of the requirements for the degree of B.Sc. in Computer Science and Information Technology be processed for evaluation.

.....
Mr. Ashok Kumar Pant
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Affiliate To Tribhuvan University

CERTIFICATE OF APPROVAL

The undersigned certify that they have read and recommended to the Department of Computer Science and Information Technology for acceptance, an internship report submitted by Mr. Hemanta Pokharel in partial fulfillment for the degree of Bachelor of Science in Computer Science & Information Technology (B.Sc.CSIT).

----- Mr. Ashok Kumar Pant Project Supervisor Chief Technical Officer Treeleaf AI Sankhamul, Lalitpur	----- Mr. Harendra Raj Bist Academic Head Department of Computer Science and IT Vedas College Jawalakhel, Lalitpur
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I am also deeply thankful to Mr. Ashok Kumar Pant, CTO at Treeleaf AI and Mr. Harendra Raj Bista, Academic Head, and Program Supervisor at Vedas College, for his unwavering encouragement, guidance, and willingness to provide his time and expertise.

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Lastly, I would like to acknowledge the contributions of all faculty members and friends who have directly or indirectly supported my project. Your suggestions and constructive criticism are highly appreciated and acknowledged.

Sincerely,

Hemanta Pokharel,

TU Exam Roll No: X

ABSTRACT

This report provides a comprehensive overview of the experiences and learning outcomes from an internship at Docsumo. It highlights key activities, projects, and responsibilities undertaken, detailing the skills acquired and challenges encountered. The report reflects on professional growth and personal development achieved through interactions with colleagues and mentors. It outlines the specific tasks and projects undertaken, as well as the tools and technologies utilized. Finally, the report includes reflections on the overall internship experience at Docsumo.

Keywords: Docsumo, key activities, projects, tools, technologies, professional growth, personal development

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LIST OF ABBREVIATIONS

AI - Artificial Intelligence

API - Application Programming Interfaces

DMS - Document Management System

ML - Machine Learning

OCR - Optical Character Recognition

OKR - Objectives and Key Results

PDF - Portable Document Format

SaaS - Software as a Service

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CHAPTER 1

INTRODUCTION

1.1 Introduction

Document processing has become a fundamental aspect of business operations across various industries. Many organizations manage an abundance of unstructured documents such as invoices, receipts, bank statements, and contracts. This proliferation of documents necessitates efficient data extraction methods. Traditional manual data entry is not only time-consuming but also error-prone, leading to inefficiencies and inaccuracies in data management. To address these challenges, Intelligent Document Processing (IDP) employs technologies such as Optical Character Recognition (OCR) and Artificial Intelligence (AI) to automate data extraction, thereby converting unstructured data into structured formats that facilitate easier analysis, reporting, and storage.

Docsumo is an intelligent document processing platform designed to meet these needs. The platform simplifies data extraction from a multitude of document formats, providing businesses with structured, validated, and accurate data. By automating the extraction workflows, Docsumo enables organizations to enhance operational efficiency, reduce manual labor, and eliminate errors associated with manual data entry. This enables a seamless flow of information across various business functions, fostering better decision-making and improving overall productivity.

The Docsumo platform handles various document types, making it a versatile tool suited for industries like finance, insurance, accounting, and healthcare. By leveraging advanced machine learning models and robust OCR technologies, Docsumo ensures that the extracted data is not only accurate but also contextually meaningful, providing insightful information that drives business value. (Docsumo, n.d., #)

1.2 Problem Statement

Quality data is crucial for businesses to flourish, but the key challenge lies in efficiently extracting and processing data from unstructured documents. Manual data entry is fraught with inefficiencies, leading to errors and data inconsistencies. In today's fast-paced business environment, organizations need to rapidly process large volumes of documents while maintaining high accuracy. To tackle this problem, companies like Docsumo provide intelligent document processing solutions that automate data extraction and validation processes.

The challenge further extends to ensuring that the extracted data is in a structured and usable format for analysis and reporting. An automated solution must handle a diverse range of document types and formats to deliver insightful and precise information, aiding businesses in activities like financial analysis, claims processing, regulatory compliance, and operational efficiency.

By integrating intelligent document processing solutions, organizations can overcome the bottlenecks associated with manual data handling, thereby accelerating business workflows and improving data quality. This fosters better decision-making and enhances the overall efficiency of business processes. (Haddad, 2021)

1.3 Objectives

The primary objective of utilizing Docsumo's intelligent document processing capabilities is to provide users with high-quality, structured data extracted from unstructured documents, facilitating various business processes. The following are the specific objectives aimed at leveraging Docsumo's capabilities:

- To build an understanding of document processing techniques and the advanced capabilities offered by Docsumo.

- To develop automated workflows that extract data from various types of documents, including invoices, receipts, and bank statements.
- To integrate Docsumo into existing business processes to enhance operational efficiency and reduce the errors associated with manual data entry.
- To analyze system requirements, prepare models, and design the architecture of an intelligent document processing system suitable for diverse business needs.
- To establish use cases that demonstrate the practical applications and benefits of intelligent document processing in different industry contexts.

1.4 Scope and Limitation

The primary scope of utilizing Docsumo is to create a comprehensive system that delivers accurate, structured data from unstructured documents. This system aims to improve operational efficiency across different business domains, such as finance, insurance, accounting, and healthcare.

1.4.1 Scopes

- **Financial Services:** Automate the extraction and processing of financial documents such as bank statements and loan applications, thereby speeding up financial analysis and reporting.
- **Insurance:** Expedite claims processing by efficiently extracting data from claim forms and supporting documents, reducing the time to settle claims and improving customer satisfaction.
- **Accounting:** Streamline data entry processes for receipts, invoices, and other financial documents, leading to more accurate and timely accounting records.
- **Healthcare:** Enhance data management in healthcare by automating the extraction of data from patient records, insurance documents, and other healthcare paperwork, thus improving operational efficiency and patient care.
- **Regulatory Compliance:** Ensure compliance with legal and regulatory requirements by accurately extracting and managing document-related data.

1.4.2 Limitations

- **Handling Diverse Formats:** While Docsumo supports a wide array of document formats, extremely diverse or highly customized formats may require additional configuration and adjustments to ensure accurate data extraction.
- **Quality of Document Scans:** The accuracy of data extraction heavily depends on the quality and clarity of the scanned documents. Poor quality scans may lead to errors and require manual intervention.
- **Integration Complexity:** Integrating Docsumo with existing legacy systems might present challenges and may necessitate custom development efforts to achieve seamless integration.
- **Accuracy in Complex Documents:** Documents with highly complex structure might be challenging to get high accuracy.

1.5 Report Organization

Report organization is an important part for report formation and understanding the report properly. It gives ideas about the structure of the report. For readability, this document is divided into several chapters and sub chapters including all details of the project. Here's brief description about all the chapters:

Chapter 1: The first chapter includes a short introduction of the project, the main problem that has been attempted to solve, the major objective of the project as well as the scope and limitation of the project.

Chapter 2: In this chapter, we add the details about the organization including the organizational hierarchy and its working domains.

Chapter 3: The third chapter includes the description of all the internship activities done during the internship period.

Chapter 4: The fourth chapter includes conclusion and the learning outcome of the entire internship period.

CHAPTER 2

ORGANIZATION DETAILS AND LITERATURE REVIEW

2.1 Introduction to the Organization

Docsumo is a leading technology company specializing in document data extraction and automation co-founded by Mr. Bikram Dahal (Nepal) and Mr. Rushabh Sheth (India). With a strong focus on cutting-edge artificial intelligence and machine learning techniques, Docsumo aims to revolutionize the way businesses handle their document processing and data extraction needs.

The company is driven by a vision to streamline manual document-based workflows, reduce errors, and improve overall operational efficiency. By harnessing the power of advanced algorithms and intelligent automation, Docsumo enables organizations to extract valuable data from documents like Invoices, Receipts, Purchase Orders, Contracts, Bank Statements, Insurance Forms, Tax Forms, KYC and more accurately and efficiently, saving both time and resources. (*Document Processing*, n.d.)

2.2 Organizational Hierarchy

The engineering and technical team at Docsumo is led by Mr. Bikram Dahal, the CTO. As the leader of the technical team, he is involved in most of the decision-making processes related to engineering and technology, taking care of the technical employees, and assigning tasks to both seasoned and new developers. On the other hand, the marketing and other organizational activities are managed by Mr. Rusheb Sheth, the CEO.

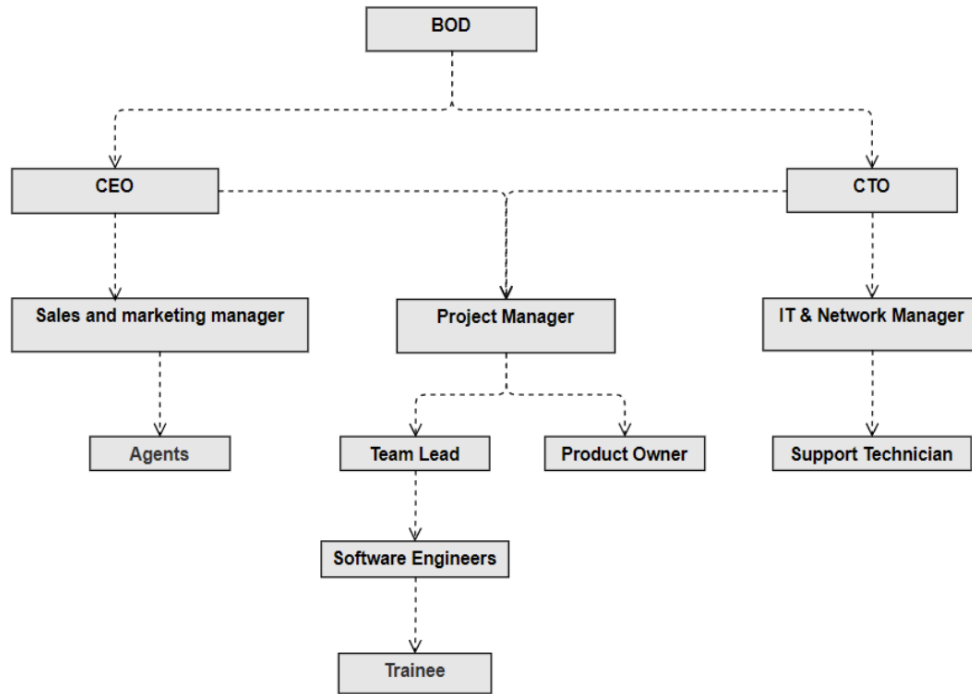


Figure 1.1: Organizational Hierarchy

2.3 Working Domain of the Organization

Docsumo offers its services in various areas related to document data extraction and automation. The primary service areas of Docsumo include:

1. **Document Data Extraction:** Docsumo specializes in extracting structured data from unstructured documents such as invoices, receipts, purchase orders, contracts, and more. Their advanced algorithms and machine learning techniques accurately extract relevant information, saving time and effort.
2. **Document Automation:** Docsumo helps businesses automate their document-centric processes, streamlining workflows and reducing manual intervention. By automating tasks like data entry, validation, and processing, organizations can improve operational efficiency and productivity.

3. **OCR (Optical Character Recognition):** Docsumo utilizes OCR technology to convert scanned documents and images into machine-readable text. This allows for easier data extraction, searchability, and integration with other systems.
4. **Data Validation and Quality Assurance:** Docsumo ensures data accuracy and quality by performing validation checks on the extracted information. This helps organizations maintain data integrity and make reliable business decisions.
5. **Integration and APIs:** Docsumo provides integration capabilities and APIs (Application Programming Interfaces) that enable seamless integration with existing systems, such as accounting software, ERP (Enterprise Resource Planning) systems, or custom applications.

2.4 Description of Intern Department/Unit

Docsumo has an effective intern department. Interns are paired with a mentor who specializes in their working domain. The mentor provides a roadmap for learning specific technologies relevant to document processing. Interns initially learn these technologies from available online resources. Following this, the mentor delves deeper into the topics and assigns tasks or minor projects to each intern. At the end of the day, interns are required to report their task completion status to their mentors.

During the internship period, various tasks and projects were assigned to the interns. The main focus was on fixing existing issues and adding new features as requested by clients. Tasks were assigned through ClickUp with deadlines, and project-related notifications and messages were managed through Slack. Upon completing the tasks, the mentor reviewed the code for complexity and efficiency. After incorporating the mentor's feedback, the code was approved and then pushed to the development branch.

2.4.1 Duration

The internship duration as per university is mention below:

Table 1: Internship Details

Internship Position	Python Developer
Time Period	March 1, 2024 - May 31, 2024
Working Days	5 days
Office Hours	9:30 AM -6:30 PM
Mentor	Mr. Suraj Shrestha

2.5 Literature Review

Docsumo leverages advanced data extraction and document management technologies to enhance business processes by enabling efficient and accurate information processing. These technologies facilitate the secure use, sharing, and interpretation of data from various document formats, thereby streamlining business operations and improving decision-making capacities.

Globally, efficient data management systems are critical for the smooth operation of various industries, including finance, insurance, and logistics. Docsumo's solutions, designed to automate data extraction from invoices, forms, and other documents, are particularly beneficial in resource-rich environments. However, these technologies also hold potential in resource-constrained settings by reducing manual workload and improving data accuracy. Docsumo's market cap indicates a growing interest in and potential for data automation technologies to improve operational efficiency, data integrity, and business decision-making through wide adoption.

The design and implementation of automated data extraction tools are important for businesses seeking to streamline operations and enhance productivity. When a document is

processed through Docsumo, an automated journey is initiated where the document is analyzed, relevant data is extracted, and the information is seamlessly integrated into existing business systems. This process involves various stages – from initial data capture and validation to integration and utilization in business workflows – each of which contributes significant data points crucial for improving operational efficiency and decision-making. The data collected and processed using Docsumo’s solutions can be utilized for performance improvement projects, anomaly detection, process optimization, and outcome analysis within the business ecosystem. (*What Is Document Processing*, n.d.)

	 Manual Entry	 Rule based software	 Intelligent Document Processing
Processing time	15+ minutes/document	2 to 5 minutes/document	30 sec to 1 min/document
Accuracy	Can yield up to 20-30% inaccuracy	85-90% accuracy	98%+ accuracy achievable
Human intervention	Completely manual	Require human-in-the-loop for processing	Human intervention occasionally needed for review/data validation
Contextual data interpretation	Yes	No	Yes
Self learning capacity	Yes	No	Yes
Complex structured documents	Can be processed	Not possible to process	Can be processed
Adaptable to different document layout	Yes	No	Yes

Figure 2: Manual extraction vs Intelligent Document processing

CHAPTER 3

INTERNSHIP ACTIVITIES

3.1 Roles and Responsibilities

As a Python Developer intern at Docsumo, the primary roles were to develop, optimize, and maintain backend functionalities of various applications. Below are the roles and responsibilities that were assigned to me:

- **Developing and Maintaining APIs:** Creating and maintaining RESTful APIs using Flask to support various front-end and mobile applications.
- **Database Management:** Utilizing MongoDB and Pymongo for database operations, ensuring efficient data storage and retrieval.
- **Containerization and Deployment:** Working with Docker to containerize applications and Kubernetes for orchestrating these containers in production environments.
- **Cloud Infrastructure:** Managing and deploying applications on Google Cloud Platform (GCP), ensuring scalability and reliability.
- **Asynchronous Task Management:** Using Celery for managing and scheduling background tasks, ensuring smooth and efficient task execution.
- **Application Monitoring and Debugging:** Setting up monitoring tools and alerts to track application performance, and debugging issues to maintain system stability.
- **Collaborating with Cross-functional Teams:** Working closely with frontend developers, data scientists, and other team members to integrate and deploy new features.
- **Documentation:** Creating and maintaining comprehensive documentation for the backend systems and processes to facilitate knowledge sharing and onboarding of new team members.

3.2 Weekly Log

Table 2: Weekly logs

Week: I

Month: I

Date	Activity
2024-03-04	• Call with HR for list of onboarding tasks • Understood Docsumo's product and its videos • Learnt about the company culture and other docs
2024-03-05	• Went through the companies case studies • Went through other recorded videos regarding Docsumo's product
2024-03-06	• Completed laptop set up in sync with company standards • Looked into companies competitors and comparing with Docsumo to create a report on it
2024-03-07	• Completed and prepared a comparison report on Docsumo and its competitors • Went through Docsumo architecture • Went through documentation under Code of conduct
2024-03-08	• Completed going through the code of conduct section • Completed reading company recommended code articles and blogs • Went through the backend structure and documentation

Week: II

Month: I

Date	Activity
2024-03-11	• Set up repositories, packages and environment • Started working on onboarding engineering QR code extraction API assignment

	• Read few code articles and design patterns
2024-03-12	• Completed the QR code extraction API onboarding engineering assignment • Went through all other code articles and design patterns
2024-03-13	• Connected with mentor to discuss the QR code extraction API task • Learned about the CI/CD and deployment pipelines • Went through internal repositories and packages
2024-03-14	• Successfully deployed the engineering task on live • Completed going through the internal packages • Attended company OKR meeting
2024-03-15	• Went through the company documentations • Set up call with every team leads to get to know the team better

Week: III

Month:I

Date	Activity
2024-03-18	• Assembled the onboarding tutorials, notebooks and references on a single repository • Started working on a bash script needed by the team to be run on VMs on cloud
2024-03-19	• Researched about the problem case and implementation for the bash script • Completed the bash script that would be used on each VMs
2024-03-20	• Attended company KSS meeting • Documented the bash script to be shared with the team • Set up backend environment and packages to work on a task on backend
2024-03-21	• Completed the backend task and sent MR for the changes

	Started working on engineering task that required to use internal tools and packages
2024-03-22	- Completed the engineering task - Built and deployed an API to complete the engineering task - Explained Gitlab CI/CD pipeline and deployment configuration to fellow interns

Week: IV

Month:I

Date	Activity
2024-03-25	- Discussed with senior team member regarding future enhancements on the bash script - Updated the script and documentation regarding it - Started working on another engineering task to integrate APIs on the product
2024-03-26	- Presented an demo regarding how to use the bash script on VMs to the ML team - Updated the bash script with new changes - Continued working on the engineering task to build and integrate API on the product
2024-03-27	- Attended biweekly OKR meeting - Completed the engineering task and deployed the API endpoints live - Started working on pre and post processing for the custom code API integration
2024-03-28	- Attended product planning meeting - Completed the pre and post processing of the custom code integration - Explored about an internal framework for the next set of task

	• Reported a bug on frontend web to the team
2024-03-29	• Started working on custom code for company's client requirements • Set up all the packages and environment needed • Call with senior team members regarding the issue and solution • Explored the codebase of the internal package

Week: I

Month: II

Date	Activity
2024-04-01	• Understood the whole architecture of the API • Experimented with codebase • Created a flowchart of the codebase
2024-04-02	• Was on sick leave
2024-04-03	• Attended company session call • Raised MR for the code changes on backend • Started working on benchmarking two implementations
2024-04-04	• Benchmarking task on hold due to reasons • Started going through the backend API codebase for the next task • Refactored the endpoints
2024-04-05	• Coordinated with the team regarding the changes requirement • Implemented changes on the codebase and tested on the product

Week: II

Month: II

Date	Activity
2024-04-08	• Coordinated with QA regarding the changes on the feature • Updated changes on feedback given on the MR
2024-04-09	• Explored another codebase for new task • Understood the endpoints which needed changes

	Started working on adding extra features on the existing MR
2024-04-10	- Completed implementing additional features and pushed the code - Continued exploring internal package for the next task - Found out what changes require to fix the issues
2024-04-11	- Implemented changes to fix the issues on the codebase - General call with engineering manager regarding the tasks - Tested out why the script was not working for another task, found the issue and solved it
2024-04-12	- Tested the implementation from the product - Experimented with the feature, to analyze what changes needs to be made for a new task - Understand the post processing part of CSV documents - Attended Company session call

Week: III

Month: II

Date	Activity
2024-04-15	- Implemented and tested the changes required for the fix - Sync with QA on the transfer context for the test - Updated the bash script to add new requirements - Discussion on how to document endpoints
2024-04-16	- Implemented feedback changes on the code - Documented the changed endpoints - Raised MRs for the tasks
2024-04-17	- Started documenting multiple endpoints on the backend - Implemented a change required by client
2024-04-18	- Product demo call - Explained CI/CD pipelines to fellow interns - Continued the backend endpoints documentation
2024-04-19	Completed documentation of the endpoints

	• Company OKR call
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Week: IV

Month: II

Date	Activity
2024-04-22	• Pushed the extra fixes on the previous task • Started working on another backend task
2024-04-23	• Discussion with lead regarding the task implementation approach • Reported a bug to FE team
2024-04-24	• Implemented the changes on backend, got feedback on the code • Started working on a fix on bug for one of the APIs
2024-04-25	• Fixed the bug on the API • Continued working on the backend task • Attended session on backend architecture
2024-04-26	• Implemented a feature for clients requirement • Completed implementation of the backend task and started documenting the changes • Session on upload documents architecture • KSS Call

Week: V

Month: II

Date	Activity
2024-04-29	• Backend session on architecture • Coordinated with FE team regarding bug and issues • Continued implementing changes required on the backend
2024-04-30	• Completed the tasks and pushed the code • Completed documentation of the endpoints • Assigned another bug issue to FE team

	● Session on API structure
2024-05-01	● Tested the changes on the product ● Reported bugs to FE team ● Discussion with seniors regarding new client requirement ● Added changes on the previous implementation
2024-05-02	● Implemented the changes after discussion with FE team and BE lead ● Fixed the bug fix ● Backend session on custom code ● Handing off the tests to QA
2024-05-03	● Handed off the task for QA completely ● Completed the documentation ● Going through the codebase for next set of task

Week: I

Month: III

Date	Activity
2024-05-06	● Went through the codebase for the context of the task ● Got the context of task, and started brainstorming the solution ● Session of GPT usage
2024-05-07	● Completed the backend task ● Documented function flow diagram of the settings ● Call with CS team regarding a new client requirement
2024-05-08	● Worked on the client project ● Had calls with CS team and senior for the implementation
2024-05-09	● Continued working on the project ● Helped an intern with his task ● Handed off bugs on the previous task to fellow intern

Week: I

Month: III

Date	Activity
2024-05-06	• Went through the codebase for the context of the task • Got the context of task, and started brainstorming the solution • Session of GPT usage
2024-05-07	• Completed the backend task • Documented function flow diagram of the settings • Call with CS team regarding a new client requirement
2024-05-08	• Worked on the client project • Had calls with CS team and senior for the implementation
2024-05-10	• Calls to explain the changes required to fellow intern • Continued working on the project to extract tables from the document

Week: II

Month: III

Date	Activity
2024-05-13	• Continued working on the client project • Sync with CS call on updates and requirements on the project • Sync with fellow intern on his update
2024-05-14	• Continued working on the client project • Onboarded fellow intern on the task as a helping hand • Pushed the first version of API • Attended Knowledge transfer session
2024-05-15	• Continued working on the client project (A lot of discussion regarding brainstorming the approach) • Knowledge transfer session
2024-05-16	• Continued working on the clients project • Reported a weird bug to the team

	• Set up clients account for demo • OKR Call
2024-05-17	• Continued working on the clients project • Discussion calls with fellow intern regarding implementation and task distribution

Week: III

Month: III

Date	Activity
2024-05-20	• Continued working with the team on clients project • Started working on another clients requirement on another API
2024-05-21	• Finished fixing the issue on client API • Continued working on the clients project and refactoring the codebase
2024-05-22	• Discussion calls with team regarding the progress • Continued working on the clients project
2024-05-23	• Wrapped up the part of clients project task • Discussion with team on remaining implementation and guiding them to it • KSS Call
2024-05-24	• Call with CS team regarding the progress on the project • Testing the changes on sample documents • Implemented changes on backend endpoint for new task

Week: IV

Month: III

Date	Activity
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2024-05-27	• Completion of the clients project and tests are done • Prepared a report on the project • Call with senior FE regarding changes on FE • Got task overview of another backend feature
2024-05-28	• Shared report with the CS team • Raised MRs to main branch • Continued brainstorming and discussion regarding the new feature on backend • Sync with FE team regarding request/response structure
2024-05-29	• Implemented the necessary changes on BE • Wrote migration scripts needed for database • Sync with CS regarding clients project update • Knowledge transfer on Afrinova • OKR Call
2024-05-30	• Pushed the API on production and setup clients account • Continued implementing changes for the new feature on backend
2024-05-31	• Tested the new changes on BE and FE • Started documenting the endpoints • Sync with fellow intern on next set remaining work on clients project • Review Call with EM

3.3 Description of Project involved during Internship

Docsumo is an advanced data extraction and document management solution designed to automate the process of capturing and processing data from various document formats such as invoices, forms, and receipts. By leveraging cutting-edge technologies like Optical Character Recognition (OCR) and Machine Learning (ML), Docsumo enables businesses to streamline their operations, reduce manual data entry errors, and enhance overall productivity. The project focuses on creating a seamless integration of extracted data into existing business systems, providing accurate and efficient information management that

supports improved decision-making and operational efficiency across various industries including finance, insurance, and logistics. Docsumo aims to revolutionize how businesses handle document-driven data, making information easily accessible and actionable. (Martinez, 2024)

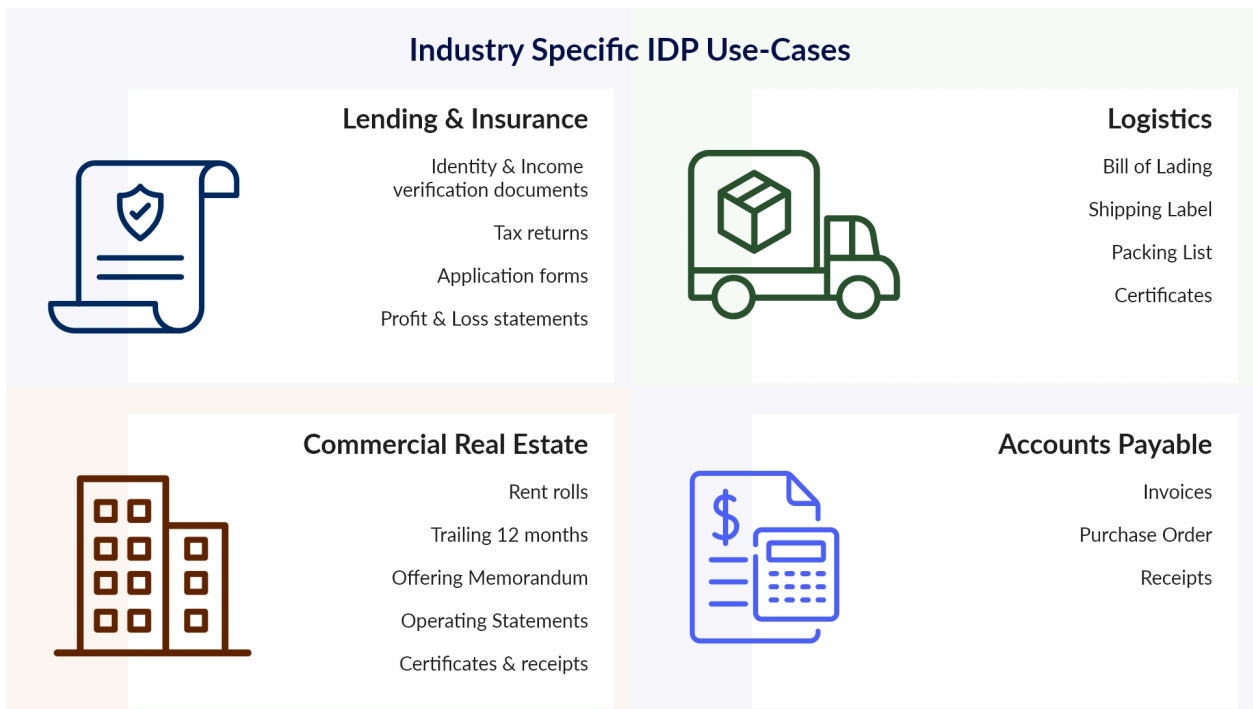


Figure 3.1: Use case of Intelligent Document Processing

3.4 Task/Activities performed

As an intern at Docsumo, the author had the opportunity to participate in a variety of tasks, projects, and activities. These included training sessions, assignments, company sessions. All of these were designed to develop our skills as software engineers, with a particular emphasis on data-related tasks. Overall, the experience was geared towards preparing them to become competent and capable professionals in the field.

3.4.1. Onboarding tasks

At first, interns had to complete a 2 weeks long onboarding tasks to understand about Docsumo, its product, how the company operates and about the company cultures. They had to analyze the competitors and make a comprehensive report on it. During the onboarding, the author also had to set up meetings with different team leads to get to know the team and how the team works better.

3.4.2. Building an extraction API

At first, interns were assigned with an engineering task, which included building an API application, which extracted and decoded QR codes and responded to the client. Also, this included setting up the CI/CD pipeline and deploying the project on cloud, making the endpoint live for everyone to use.

3.4.3 Knowledge Sharing Sessions

At Docsumo, weekly sessions are held regarding the product, system, and backend design to inform the team about the operations and developments in each part of the product. These sessions are very helpful, providing insights into the overall architecture of Docsumo.

3.4.4 Project Management Activities

Project management at Docsumo is set up using ClickUp, where the team learns about the company's agile/scrum development process and the creation and meaning of tickets. Since ticket status is very important, updating task descriptions, due dates, status, and priority is given utmost importance.

3.4.5 Engineering tasks

During the internship at Docsumo, the author had the opportunity to work on a diverse set of tasks, primarily focusing on building the backend for their innovative product. One of the main responsibilities was to develop APIs that could efficiently extract tables and key values from various types of documents, catering to the specific requirements of clients. To achieve

this, a wide range of technologies and frameworks were leveraged, including Flask, Redis, Celery, OpenCV, OCR (Optical Character Recognition), and DataFrames.

Additionally, there was the opportunity to work with cloud services such as Google Cloud Platform (GCP) to deploy and manage applications effectively. Valuable experience in containerization was gained using Docker, ensuring seamless deployment and scalability. As part of the learning journey, there was an exploration into Kubernetes, facilitating the orchestration of containerized applications and streamlining the deployment process.

Throughout the internship, active collaboration with the team was essential to optimize the performance and reliability of the backend system, contributing to the overall growth and success of the product. The experience at Docsumo has been immensely enriching, allowing the strengthening of Python skills and gaining valuable insights into cutting-edge technologies within the domain of document processing and data extraction.

CHAPTER 4

Conclusion and Learning Outcomes

4.1 Conclusion

The internship at Docsumo has been incredibly rewarding and transformative. Valuable insights were gained, essential skills developed, and significant personal and professional growth achieved. Working alongside experienced professionals, collaborating on meaningful projects, and receiving guidance from mentor Mr. Suraj Shrestha greatly contributed to overall learning and growth.

Throughout this internship, the importance of adaptability, effective communication, time management, and embracing feedback became evident. These lessons are crucial for future endeavors. Additionally, the experience of working in a collaborative team environment reinforced the significance of teamwork and collective efforts.

Gratitude is extended to college internship coordinators for their support in securing this valuable opportunity and guiding throughout the process. Their coordination and advice were instrumental in making this internship a success.

Sincere thanks are extended to everyone who contributed to this internship journey – mentor, colleagues, college internship coordinators, family, and friends for their unwavering support. Their guidance, encouragement, and belief in abilities have been invaluable.

Overall, this internship has provided a solid foundation for future career aspirations and instilled confidence and enthusiasm. The experiences, lessons, and connections gained will be applied and built upon to continue personal and professional growth.

4.2 Learning Outcome

During the internship at Docsumo, the opportunity arose to immerse in various technologies and processes integral to the company's operations in the document AI industry. Here are the key learning outcomes from the experience:

Hands-on experience with Flask, a micro web framework in Python, enabled efficient development and deployment of web applications. Proficiency was gained in using MongoDB and Pymongo for managing and querying databases, crucial for handling the vast amounts of data processed by Docsumo.

Working with Docker and Kubernetes significantly enhanced understanding of containerization and orchestration. Skills were developed in packaging applications with Docker and managing their deployment, scaling, and operations using Kubernetes, ensuring modern deployment practices and application scalability and reliability. (Ellingwood, 2018)

Familiarity with Google Cloud Platform (GCP) provided insights into various cloud services and infrastructure management. Exposure to Gunicorn, a Python WSGI HTTP server for UNIX, improved skills in deploying and managing Python applications in production environments.

The internship provided a deep dive into document extraction technologies, learning to extract and process data from various document formats. This included working with Redis for caching, which significantly improved the performance and efficiency of data retrieval processes.

Experience was gained with Celery tasks and queues, essential for managing asynchronous tasks and ensuring that time-consuming processes do not block the main application. Additionally, setting up alerts and monitoring systems was crucial for smooth operation and timely issue detection within the system.

Furthermore, the internship facilitated the development of several interpersonal and professional skills:

- **Problem-solving Skills:** Enhanced ability to tackle challenges independently and devise innovative solutions.
- **Stress Management:** Maintained productivity and performance under pressure, managing deadlines effectively.
- **Teamwork:** Collaboration with team members enhanced effectiveness within a group.
- **Communication Skills:** Improved ability to communicate technical concepts clearly and effectively.
- **Corporate Environment Adaptation:** Adapted to the dynamics and expectations of a professional setting.
- **Integration of Financial and Technological Concepts:** Gained understanding of the intersection of financial operations with advanced technologies.

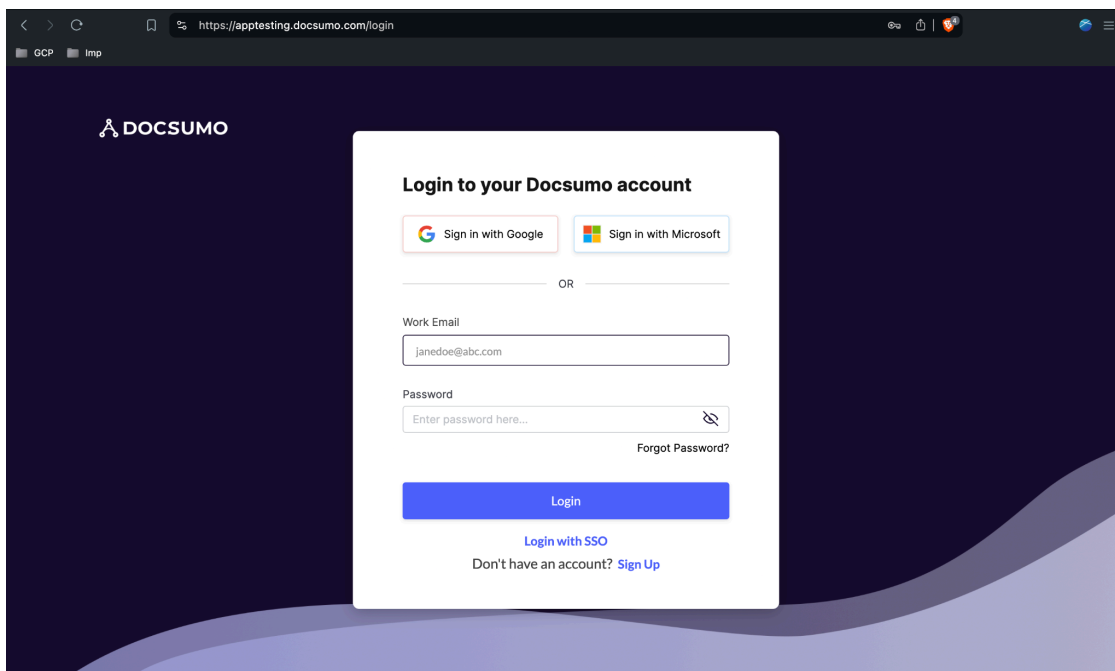
Overall, the internship at Docsumo expanded technical knowledge and skills, equipping with the ability to operate efficiently in a professional and collaborative environment.

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Appendices

Login/signup page



The screenshot shows a web browser window with the URL `https://apptesting.docsumo.com/login`. The page features a dark purple background with a white login/signup form in the center. The form is titled "Login to your Docsumo account" and includes two social login buttons: "Sign in with Google" and "Sign in with Microsoft". Below these is a horizontal line with the text "OR". The form then has a "Work Email" field with the placeholder text "janedoe@abc.com" and a "Password" field with the placeholder text "Enter password here...". A "Forgot Password?" link is located to the right of the password field. A blue "Login" button is positioned below the password field. At the bottom of the form, there is a link "Login with SSO" and a link "Don't have an account? Sign Up".

DOC SUMO

Login to your Docsumo account

[Sign in with Google](#) [Sign in with Microsoft](#)

OR

Work Email

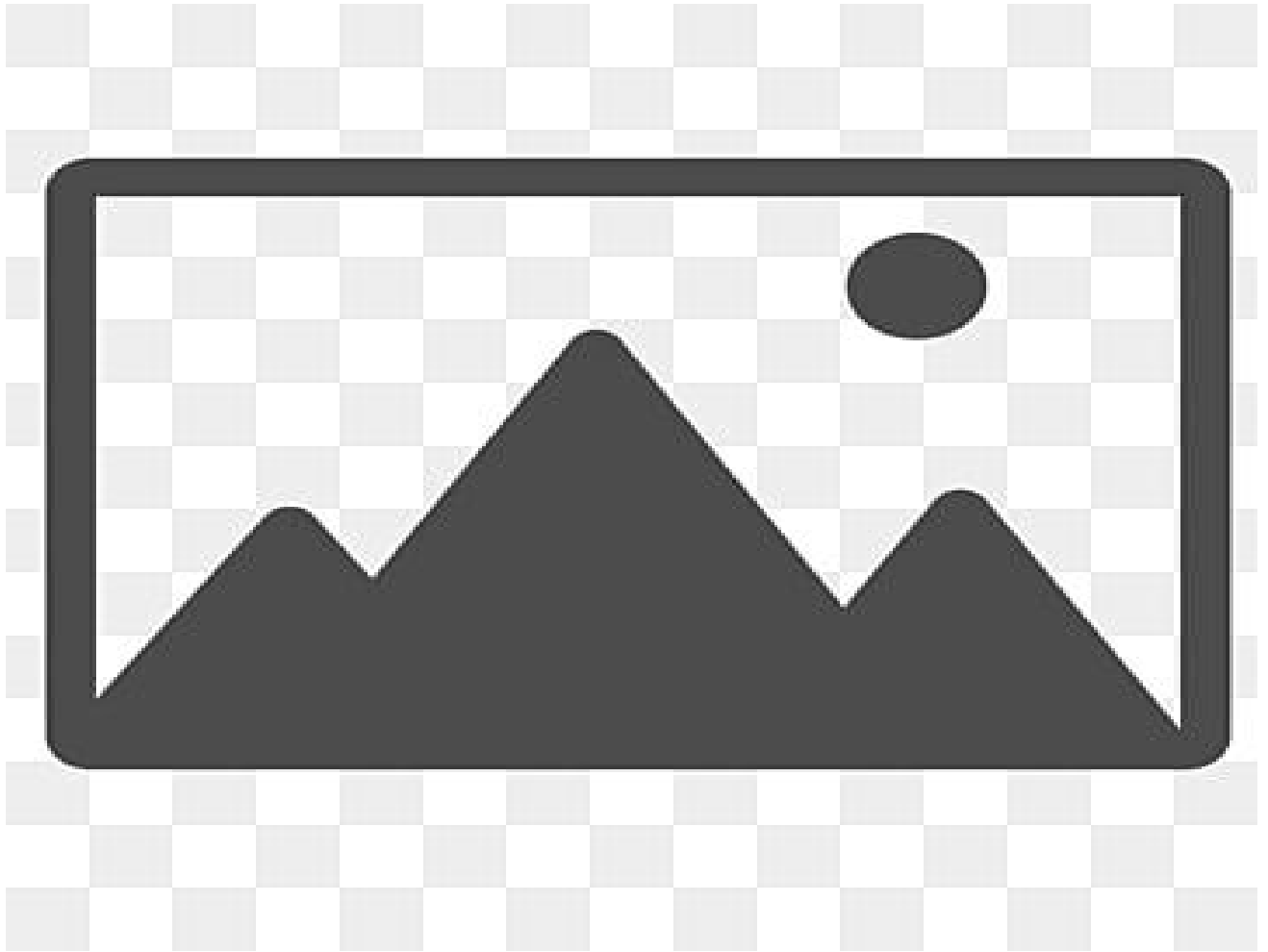
Password

[Forgot Password?](#)

[Login](#)

[Login with SSO](#)

Don't have an account? [Sign Up](#)



Home screen (**Add more images of your work**)

Special thanks to **Hemanta Pokharel**

Website: <https://hemantapkh.com/>

Github: <https://github.com/hemantapkh>